Excercise 2 - SQL foundations

1 SELECT distinct department
From students;

Students

depart
IT
HR
Finance

2 SELECT department,
AVG(average age) AS total age
From students
GROUP BY department

Students

depart	Age
IT	20,5
HR	22
Finance	23

3 SELECT department
COUNT (student) AS student count
FROM students
Group by department
HAVING (students) > 1

Students

depart	student
IT	2
HR	2

4 SELECT Student ID, Name, Age, department FROM students WHERE age BETWEEN 21 AND 26; OR SELECT *

Students

Student ID	Name	Age	depart
2	Bob	22	HR
3	Charlie	21	IT
4	Dianna	23	Finance
5	Eve	22	HR

5 SELECT*
FROM students
WHERE Department IN ('IT', 'HR')
AND Age > 21

Styclents

Student ID	name	age	depart
2	Bob	22	HR
5	Eve	22	HR

6 SELECT department,
Sum(credits) AS total credit
FROM courses
GROUP BY department
HAVING Sum(credits) > 5

Cause

department	credits
IT	11

7 SELECT ID,
name,
department,
credit
From courses
WHERE NOT = 4;

course

ID	name	depart	credits
101	SQL Basic	IT	3
104	Excel	Finance	2
105	Stats	HR	3

8 SELECT ID
name
credits

Course

ID	name	credits
102	Python	4
103	data Science	4
101	SQL Basics	3

9 SELECT Max grade
Min grade
Avg grade
FROM enrollments
course enrollment

Max grade	Min grade	Avg grade
99	78	

10 SELECT ID
COUNT enrollment
from enrollments
order By Course ID

Course ID → 5

11 SELECT department
Sum (Salary) AS total salary
Sum (bonus) AS total bonus
FROM salaries
Group by department

salaries

department	total salary	total bonus
IT	122 000	10 500
HR	109 000	7500
Finance	70 000	6 000

12 SELECT department
AVG (salary) AS avg_salary
FROM salaries
GROUP BY department
HAVING AVG (salary) > 55 000

salaries

department	avg salary
IT	61 000
Finance	70 000

13 SELECT employee id
name
salary
bonus
, salary+bonus AS total compensation
FROM salaries
WHERE ~~total~~ salary+bonus > 60 000

salaries

ID	name	salary	bonus	Total compensation
1	tom	60 000	5 000	65 000
2	spike	70 000	6 000	76 000
3	tyke	82 000	5500	87 500

14 SELECT department
SUM(Budget) AS total budget
AVG(Budget) AS Avg budget
FROM projects
Group by department
HAVING AVG(Budget) > 70 000

department	total budget	AVG budget
IT	270000	135 000
finance	80000	80000.

15 SELECT project id
project name
department
budget
FROM projects
WHERE budget BETWEEN 50 000 AND 120 000
AND department NOT Marketing.

projects

project id	project name	department	budget
1	AI APP	IT	
2	Payroll System	finance	
3	Dashboard		
4	website		
5.	HR Portal		

project id	project name	department	budget
1	AI App	IT	120 000
2	Payroll	Finance	80 000
5	HR portal	HR	50 000.